

Exercise and worked solution

Estimation & hypothesis testing

Four parts to move from theory to data: the precision of an estimate, correcting the variance bias, running a hypothesis test, and reading a p-value honestly. Work through each part before reading the solution.

Part A · Precision of an estimate

PROBLEM

A monthly return has an estimated standard deviation of $\sigma = 16.5\%$ over $n = 68$ observations. (a) Compute the standard error of the mean. (b) Explain why it differs from the standard deviation.

SOLUTION

1. (a) Standard error = $\sigma/\sqrt{n} = 16.5 / \sqrt{68} = 16.5 / 8.246 \approx \mathbf{2.00\%}$.
2. (b) The **standard deviation** (16.5%) measures the dispersion of the data: it is fixed and does not depend on n . The **standard error** (2.00%) measures the uncertainty on the estimated mean: it decreases in $1/\sqrt{n}$. In other words, we know the mean of a large sample far more precisely (2%) than the raw volatility (16.5%) suggests.

Part B · Unbiased variance

PROBLEM

Over $n = 10$ observations, the sum of squared deviations from the mean is 180. Give the biased estimator ($\div n$) and the unbiased estimator ($\div n-1$), and the size of the correction.

SOLUTION

1. Biased: $\sigma^2 = 180 / 10 = \mathbf{18}$.
2. Unbiased: $s^2 = 180 / 9 = \mathbf{20}$.
3. The correction is $n/(n-1) = 10/9 \approx 1.11$: dividing by n understates the variance by 11% here. The bias comes from centring deviations on $\hat{\mu}$ (computed from the same data), which consumes a degree of freedom. It becomes negligible once $n \geq 30$.

Part C · Running a test

PROBLEM

An index's mean excess return is $\hat{\mu} = 7.62\%$, with $\sigma = 16.5\%$ and $n = 68$. Test $H_0: \mu = 0$ against $H_1: \mu \neq 0$ at 5%, and give the 95% confidence interval.

SOLUTION

1. Standard error = $16.5 / \sqrt{68} \approx 2.00\%$.
2. Statistic: $T = (\hat{\mu} - 0) / \text{standard error} = 7.62 / 2.00 \approx \mathbf{3.8}$.
3. Decision: $3.8 > 1.96$ (two-sided 5% critical value) $\rightarrow$ **reject H_0** . The mean return is significantly different from zero (p-value $\approx 0.01\%$).
4. 95% confidence interval: $\hat{\mu} \pm 1.96 \times \text{standard error} = 7.62 \pm 1.96 \times 2.00 = 7.62 \pm 3.92 = \mathbf{[3.70\%; 11.54\%]}$. It excludes 0, which gives exactly the rejection decision.

Part D · Annualisation and reading the p-value

PROBLEM

(a) A daily return has a mean of 0.03% and a volatility of 1.0%. Annualise them (252 days). (b) A manager gets a p-value of 3% testing whether they beat their index. Can they conclude “I have a 97% chance of beating the index”?

SOLUTION

1. (a) Annualised mean: $0.03\% \times 252 = \mathbf{7.56\%}$. Annualised volatility: $1.0\% \times \sqrt{252} \approx 1.0\% \times 15.87 = \mathbf{15.87\%}$. The mean scales in s , volatility in $\sqrt{s}$ (square-root-of-time rule, valid under time independence).
2. (b) **No.** The 3% p-value means: *if* the manager did not beat the index (H_0 true), there would be only a 3% chance of observing an outperformance at least as large. This is neither $\Pr(H_0 \text{ true})$ nor $\Pr(\text{they beat the index})$. We reject H_0 at 5% (since $3\% < 5\%$) and conclude, more cautiously, that the outperformance is statistically significant. Confusing the p-value with the hypothesis probability is the most common interpretation error.

Key takeaways: the standard error ($\sigma/\sqrt{n}$) states precision and decreases with n ; correct the variance bias with $n-1$; a test compares $T = (\hat{\mu} - \mu_0)/(\sigma/\sqrt{n})$ to a critical value; and a p-value is never the probability that the null hypothesis is true.